

PROJECT CHARTER

EEG-Based Clinical Decision Support Tool

Product Feasibility & Business Analysis Study

Author: Bhumi Nagar | Domain: Healthcare AI / Psychiatric Clinical Decision Support |

Type: Feasibility Study & BA Portfolio Project

IMPORTANT: This project is a proof-of-concept feasibility study and does not represent a clinically deployable diagnostic system. All findings are exploratory and intended for business analysis and portfolio demonstration purposes only.

1. Project Overview

This project evaluates the feasibility of an AI-assisted clinical decision support concept for schizophrenia risk classification using EEG-derived healthcare data. The initiative combines machine-learning validation, business analysis, and healthcare workflow assessment to assess whether predictive analytics could support early-stage psychiatric screening in a clinically responsible manner.

The core premise is that schizophrenia — a chronic psychiatric disorder — exhibits measurable neurophysiological signatures in NREM sleep EEG data before clinical severity is reached. If these patterns can be reliably detected, they may serve as objective biomarkers to support psychiatric decision-making workflows.

The project integrates:

- Business Analysis (BA) — requirements, stakeholder workflows, feasibility evaluation
- Healthcare Analytics — EEG-derived biomarker identification and interpretation
- Machine Learning Validation — classification model development and evaluation
- Clinical Workflow Evaluation — As-Is and To-Be process modeling
- Responsible AI Considerations — deployment constraints, regulatory awareness, ethical risks

2. Business Problem

Psychiatric disorders such as schizophrenia often involve delayed diagnosis and subjective clinical evaluation processes. Traditional assessment methods rely heavily on behavioral observation, clinical interviews, and longitudinal evaluation — approaches that may increase diagnostic variability, delay early intervention, and limit scalability across diverse clinical settings.

This project explores a focused question: can EEG-derived neurophysiological patterns, analyzed through machine learning, contribute to a structured decision-support workflow for mental health clinicians — particularly in detecting early-stage risk before chronic severity is established?

3. Project Objectives

1. Evaluate the technical feasibility of ML-based schizophrenia classification using EEG-derived NREM sleep features
2. Assess whether predictive analytics can meaningfully support clinical decision-making workflows
3. Analyze operational, analytical, and deployment limitations associated with healthcare AI systems
4. Develop structured BA artifacts, including workflow models, BRD, feasibility evaluation, and stakeholder analysis
5. Apply responsible AI and regulatory frameworks (FDA SaMD guidance) to assess real-world deployment readiness
6. Produce a structured go/no-go feasibility recommendation based on ML study evidence

4. Scope

Included

- EEG-derived healthcare dataset analysis (130 subjects, ~2,894 features)
- Machine learning model development, comparison, and evaluation
- Feature selection and statistical validation (ANOVA SelectKBest)
- As-Is and To-Be clinical workflow modeling
- Clinical decision support feasibility assessment
- Business analysis documentation (BRD, stakeholder analysis, risk assessment)
- Regulatory landscape review — FDA SaMD classification framework
- Responsible AI considerations for psychiatric healthcare contexts

Excluded

- Real-time clinical deployment or live hospital integration
- FDA approval processes or formal regulatory submission
- Patient-facing implementation or direct clinical use
- Raw EEG signal processing (pre-extracted features used)
- External clinical validation with independent datasets
- Deep learning or neural architecture model development
- Drug, treatment, or dosage recommendation systems

5. Key Stakeholders

Stakeholder	Role	Primary Interest
Psychiatrists / Clinicians	Primary End Users	Early detection support, structured decision workflow
Patients	Indirect Beneficiaries	Earlier diagnosis, improved long-term outcomes
Hospital / Clinic Administration	Deployment Decision-Makers	Cost efficiency, compliance, operational fit
Healthcare IT Teams	Integration Owners	EHR compatibility, data infrastructure, and security

Stakeholder	Role	Primary Interest
Regulatory Bodies (FDA)	Compliance Authority	SaMD classification, patient safety, standards
Data Scientists / ML Engineers	Technical Builders	Model performance, maintainability, scalability
Bhumi Nagar (Project Author)	BA Lead / Analyst	Feasibility evaluation, BA documentation, portfolio

6. Project Deliverables

- ML Validation Notebook — Python pipeline with full preprocessing, model training, and evaluation
- Project Charter — This document
- Business Requirements Document (BRD) — Functional and non-functional requirements, constraints
- As-Is vs To-Be Workflow Models — Clinical process diagrams (Lucidchart)
- Feasibility & Go/No-Go Assessment — Evidence-based deployment recommendation
- AI Deployment Constraints Document — Infrastructure, HIPAA, EHR integration considerations
- Responsible AI Considerations Document — Bias, explainability, patient safety framework
- GitHub Repository — All artifacts organized and publicly documented

7. Project Phases

Phase	Name	Key Activities
Phase 1	Technical Validation	Data acquisition, cleaning, ML modeling, cross-validation, evaluation
Phase 2	BA Documentation	BRD, stakeholder analysis, requirements definition, workflow modeling
Phase 3	Feasibility Assessment	Regulatory review, deployment constraints, go/no-go evaluation
Phase 4	Responsible AI	Bias analysis, ethical framework, patient safety considerations
Phase 5	Repository & Presentation	GitHub packaging, README, visuals, presentation artifacts

8. Risks & Constraints

Risk / Constraint	Impact	Mitigation / Notes
Small sample size (130 subjects)	High	Cross-validation applied; results treated as exploratory, not clinical-grade
High-dimensional feature space (~2,894 features)	High	ANOVA SelectKBest applied; K=100 features selected to reduce dimensionality
Risk of model overfitting	High	Stratified 5-fold CV used; CV AUC std monitored; limitations explicitly documented
No external validation dataset	High	Flagged as a critical limitation, Wave 2 replication by the original authors noted
Pre-extracted EEG features only (not raw signals)	Medium	Acknowledged as a scope constraint, raw EEG processing is deferred to future work
Regulatory & ethical constraints for healthcare AI	High	FDA SaMD framework reviewed; deployment classified as not ready in go/no-go
Data privacy / HIPAA compliance	High	An anonymized dataset was used; full HIPAA compliance is required for any real deployment
Model bias across demographics (age, sex)	Medium	Flagged in the Responsible AI document, a diverse dataset is required before deployment

9. Success Criteria

Criterion	Target	Status
ML model achieves AUC ≥ 0.80 on held-out test set	AUC ≥ 0.80	☑ SVM: 0.896
Methodological rigor: leakage-safe pipeline and cross-validation	No data leakage	☑ Achieved
Interpretable evaluation metrics produced (ROC-AUC, F1, confusion matrix)	All metrics present	☑ Achieved
Business requirements clearly documented and traceable	BRD complete	☑ Achieved
Regulatory and responsible AI considerations are explicitly addressed	Documented	☑ Achieved
Go/no-go recommendation supported by structured evidence	Assessment complete	☑ Achieved
All artifacts are published in a clean, navigable GitHub repository	Repo complete	☑ Achieved

10. Recommendation Summary

The feasibility study demonstrated promising classification capability using EEG-derived neurophysiological features. The SVM model achieved an AUC of 0.896 and approximately 84% classification accuracy on a held-out test set, with consistent performance across 5-fold cross-validation (mean AUC: 0.835 ± 0.105).

However, significant limitations remain in sample size, external validation, and deployment readiness. The project supports continued exploration of AI-assisted clinical decision-support workflows, while emphasizing the need for:

- Larger, more diverse clinical datasets for model generalization
- External validation on independent patient cohorts
- Responsible AI governance frameworks before clinical integration
- Regulatory alignment with FDA SaMD classification requirements
- HIPAA-compliant data infrastructure for any real-world deployment

The current study is best classified as a structured feasibility demonstration with a conditional go recommendation — contingent on resolving the identified technical, clinical, and regulatory gaps.

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